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Title:

Testsystem for the RC701/RC702
Version 1.1
User's Guide

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Abstract: This manual describes the testsystem and the testprograms for the RC701/RC702.

The testprograms described in this manual are: the ram test, the ram controller test, the dma test, the ctc test, the fdc test, the pio test, the sio test, the fdd test, the wdc test, the fdd reliability test, the crt test, and the wdd reliability test.

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1. INTRODUCTION 1.

This manual describes the diagnostic test programs for the RC700 microcomputer.

The test programs are testing the basic functions of the different parts of the hardware in the computer.

The sequence of the different test programs in the test system is organized with rising complexity. As far as possible, no part of the hardware is used before it is tested.

If an error appears then make some retries to see, if the error is permanent.

1.1 Load and Start 1.1

The test system is delivered on two flexible disks, a minidiskette and a maxidiskette, and in two PROM versions, a 4K PROM and as 2 x 2K PROM's.

Starting the test system from flexible disk:

- insert the diskette in the drive, push reset button and the tests start automatically.

Starting the test system situated in PROM's:

PROM system 1 (2 x 2K PROM)

These two PROM's are to be used with MIC702 and MIC703.

Turn off the power, remove the hood, install PROM (ROB361) in PROGRAMABLE ROM 0 socket and PROM (ROB362) in PROGRAMABLE ROM 1 socket. Turn on the power again. RESET will start the tests automatically.

PROM system 2 (4K PROM)

This PROM is to be used with MIC703, if print modif is made, and MIC704.

Turn off the power, remove the hood, install PROM (ROB363) in PROGRAMABLE ROM 0 socket. Turn on the power again. RESET will start the tests automatically.

Due to the limited storage capacity in the PROM's, only some of the tests are available. It is the tests which have connection to the flexible disk drives. The tests are: RAM test, DMA test, CTC test and FDD test.

2. THE TESTS 2.

The testsystem described in this manual consists of 12 tests. The tests are ran under control of a testrouter, which has the purpose of administering the mode in which a particular test is ran. The ten tests are:

Memory test
Memory refresh test
DMA test
CTC test
FDC test
PIO test
FDD test
SIO test
WDC test
FDD reliability test
CRT test
WDD reliability test

3. THE TESTROUTER 3.

The testrouter is a central program which main purpose is to compute the address of the next test in the sequence. The address of the next test is derived from a variable holding the present testnumber and normally incremented by one. Every time a test has finished and is not in looping mode, the testrouter is entered.

3.1 Switch Parameters 3.1

The variable holding the test number also contains four switch bits, by which the testrouter decides how to administer the tests.

7	6	5	4	3	2	1	0
				test No			

h	l	k	s
a	o	e	u
l	o	y	p
t	p	b	r
		.	.
		o	p
		n	r
			i
			n
			t

halt: 0: halt on error
1: proceed though error

loop: 0: sequential, big loop of all tests
1: looping in the present test

keyb on: 0: the system has not been informed about an installed keyboard
1: a key on the keyboard has been struck at least once

supr print: 0: messages are printed on the display
1: no messages are printed (usable for fast looping)

Initial values of the switch parameters are all zeroes.

3.2 Keyboard Management 3.2

To enter the keyboard management is only possible when a strucked key has informed the system that a keyboard is connected. When this has happened, the testrouter answers with the valid capital letters and digits.

Valid keys are as follows:

H: set halt bit to 0
R: set halt bit to 1
L: set loop bit to 1
G: set loop bit to 0
S: set suppress print bit to 1
P: set suppress print bit to 0
(Please note that capital letters are used)

<esc>: will stop execution.

This is also the fact for any other key. Striking the <return> key will have the test system reentering the looping or running state.

Numbers between 0-F will insert a new test number into the variable.

All other keys will give no response.

If one for example wants to loop in test 6 and not go into a HALT state if error, then strike the keys R, L, 6 (not necessarily this sequence). Furthermore if one wants to test fast for measuring purpose, one can suppress print by typing S. Suppress print means that nothing from now on will be written into the screen-buffer.

Relationship between the test numbers and actual tests is as follows in the diskette versions:

Test No	Test name	Test loop
0	RAM test	yes
1	Memory refresh test	yes
2	DMA test	yes
3	CTC test	yes
4	FDC test	yes
5	PIO test	yes
6	SIO test	yes
7	FDD test	
8	WDC test	See chapter 10 and 11
9	WDC test **)	Have to be selected by
A	FDD reliability test	the testnumber
B	CRT test	
C	WDD reliability test	
D	not used yet	
E	not used yet	
F	not used yet	

**) Press the key "9" will force the testsystem to perform the WDC test, see chapter 12.

Relationship between the test numbers and actual tests is as follows in PROM versions:

Test No	Test name	Test loop
0	PROM checksum and RAM test	yes
1	DMA test	yes
2	CTC test	yes
3	FDD test	See chapter 10
4	not used	
5	not used	
6	not used	
7	not used	
8	not used	
9	not used	
A	not used	
B	not used	
C	not used	
D	not used	
E	not used	
F	not used	

The test numbers not used yet will force the test system to return to test number 0.

3.3 Output 3.3

The testrouter will respond with some output. This is a version number and a test number. It also responds with the state of the test. This could be either running, stopped, looping or halted.

3.4 Specialities 3.4

The testrouter has a waiting point of 3 seconds when entered from the test number 0 to give the user time to key in some input to change parameters.

4. THE MEMORY TEST 4.

The memory test consists of two tests. A PROM checksum test and a RAM test.

4.1 PROM Checksum Test 4.1

A check on the contents of the PROM (containing tests) is performed by adding the contents of all locations in the PROM and checking that the result is FF (Hex).

If a difference from FF (Hex) is found, an attempt is made to write an error message on the first line of the display.

The error message has the following layout:

<RC700 TESTSYSTEM chksum err>

Note that when loaded from flexible discs, the checksum is checked on the loaded image.

4.2 RAM Test 4.2

The memory test thereafter performs a test of the dynamic RAM memory. Of course, all memory cells are tested by the memory test (all variables are kept in CPU registers).

First the upper part (addresses higher than the last PROM address) of the memory is tested. If this was found OK, the test is moved to this memory area and the memory space shaded by PROM is tested.

The test pattern for the dynamic RAM memory consisting of chips of 1 bit x 16 k is three times 00 followed by three times FF (Hex). When all memory cells have been tested, they are again tested with the inversed pattern. This means that all bits are tested for "zero" and "one" insertion. It is the most convenient

pattern for discovering addressing errors because this modulus 3 pattern will not be repeated equivalent in a higher modulus address.

If an error occurs, a message will be written on the first line of the display. The layout is the following:

<RC700 TESTSYSTEM mem err ha la ex re>

where "ha" is high address, "la" is low address, "ex" is expected value and "re" is received value. All numbers are in hexadecimal notation. (To find any defective chip, consult fig. 1).

When both the PROM checksum test and the RAM memory test are terminated, an attempt is made to write on the display (whether there has been an error or not). This is done as simple as possible without any standard program (without interrupt service). On the first 17 positions of the first line, the identification of the system type will be written followed by a possible error message.

The total turn around time for the RAM-test is 7.5 seconds.

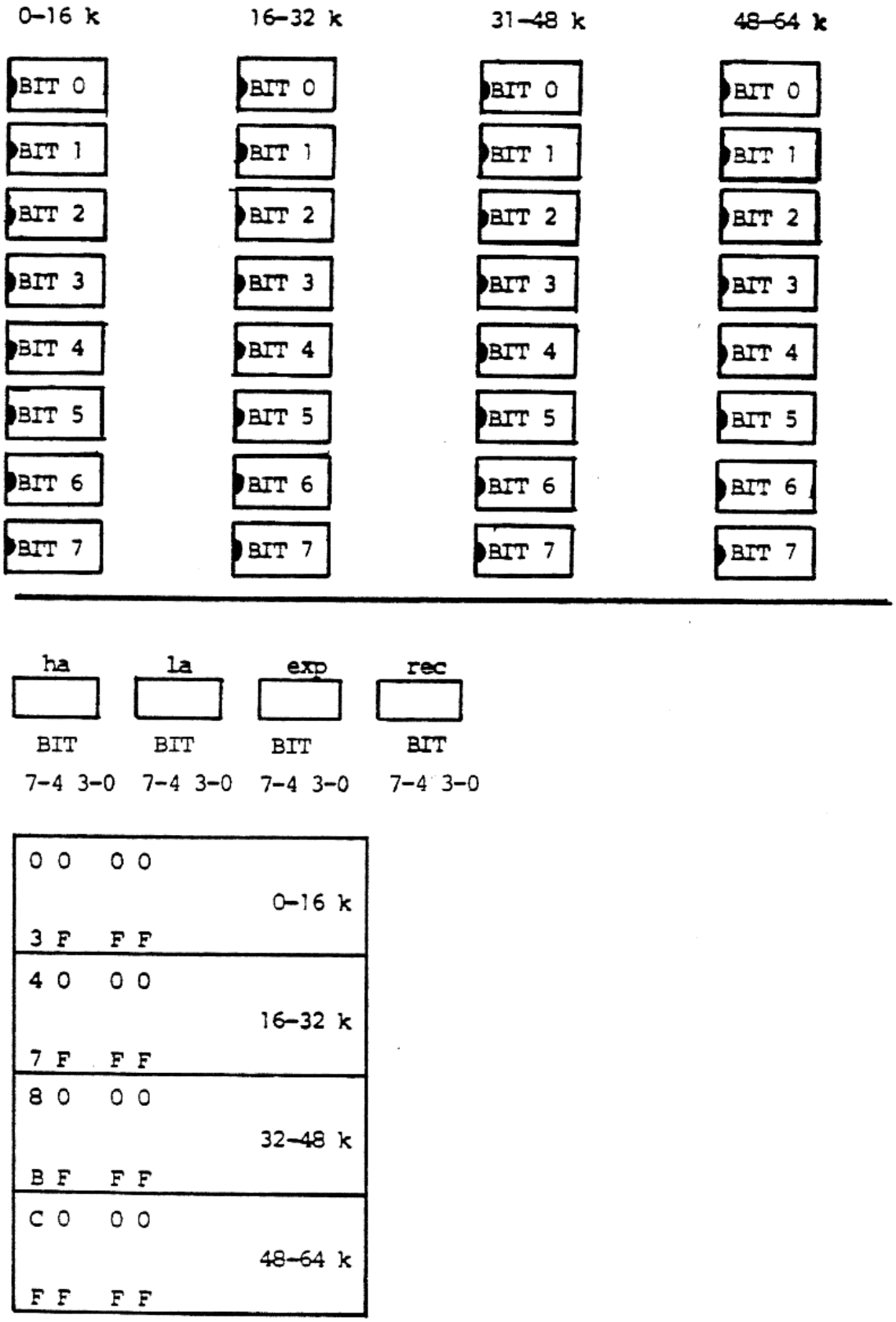

Figure 1: Layout of 64K RAM memory.

5. MEMORY REFRESH TEST 1 5.

The dynamic memory refresh test is a test which verifies the function of the memory controller chip.

It writes a pattern in memory consisting of an XOR of high and low address part. The pattern is written from the memory address 8000H until the hexadecimal address: F000H (where the display image starts).

When the pattern has been written, the test waits for 5 seconds in a waiting loop before it performs a check of data.

The main purpose of this test is to discover modification of data happened in the delay time, due to malfunction of the refresh counting in the memory controller.

Possible error message is:

<data modified in byte xx xx exp: xx rec: xx>

Output is placed on the fourth line of the display.

6. DMA TEST 2. 6.

The DMA test loop is testing DMA transfers between channel 0 and channel 1. This is done as a memory to memory transport. Channel 1 is receiving and channel 0 is transmitting.

The transmitted pattern is a buffer of 1 k containing a counting pattern. The pattern is as follows: 00 FF FE FD etc. repeated 4 times.

When the transport is finished, the receiving buffer is checked against the transmitted buffer byte by byte.

The DMA test will write its messages on the fifth line of the display.

Apart from the identification of the test, the possible messages from the DMA test are:

<OK>
<TC timeout 200 ms>
<data error, byte no: xx xx exp: xx rec: xx>

All numbers are in hexadecimal notation. "TC timeout ms" shows that the terminal count bit for channel 1 in the DMA status register has not been set within 200 ms. and that the transport is therefore not successful.

If the screeen is blanked or if the errormessages continue to appear after several retries the DMA-controller should be replaced.

7. CTC TEST 3 7.

This program is testing the counter timer circuit which is used for baud rate generator and as interrupt circuit for the CRT and the FDC controller.

It is tested that the circuit will generate interrupt and that the vector (interrupt address) is correct.

The four channels 0, 1, 2 and 3 are tested. Channels 0 and 1 are tested in counter mode, counting on the fixed input clock giving interrupt after approx. 423 µs. Channels 2 and 3 are tested in timer mode. The timer is for channel 2 started by the interrupt signal from the CRT controller. For channel 3 the timing is started automatically.

The test is based on a timeout loop, so it is checked if the interrupt was received within a specified time (300 ms.). It is also checked that only the specified channel interrupts.

The test can end up with 2 different error messages:

<illegal interrupt, port: xx>
meaning that another channel than the specified has interrupted.

<no interrupt, ch:>
meaning that the test has timed out before interrupt was received.

Texts will be written on the sixth line of the display.

8. FDC TEST 4/ 8.

The FDC test is a small test loop included in the big sequential loop of tests.

When the test is entered, it is checked that the main status register of the FDC has bit 7 set to indicate that the controller is ready.

After that, an invalid command is sent to the controller to see if it responds correctly.

The following three error messages could appear:

<not ready receive-transmit>:
bit 7 of the main status register was not set when entering the test.

<wrong data-direction>:
bit 6 of the main status register has wrong polarity.
Push the RESET button and try again.

<fault stat. reg.>:
status register 0 should indicate an invalid command (bit 7 = 1, bit 6 = 0).

Texts will be written on the seventh line of the display.

9. PIO TEST 5 9.

To run this test it is necessary to use the testcable CBL936, where plug J3 of the cable is installed in plug J1003 (PARALLEL OUT) of the computer, and plug J4 of the cable is installed in plug J1004 (KEYBOARD) of the computer.

If no testcable is installed the testsystem will respond with received and expected data and "pio_test aborted".

To test if a testcable is installed the program transmits the byte E5 (hexadecimal) at the parallel out port and checks the keyboard port. If the keyboard port holds the byte E5 it is assumed that the cable is installed, otherwise not.

If the test is ran in looping a rolling counting pattern of 256 bytes is transmitted and checked against the received pattern.

The test can end up with 3 different error messages:

<no interrupt from port A>
means that an interrupt from the keyboard port has not occurred within 20 T-cycles

<channel B interrupt>
means that an interrupt from the parallel out port has interrupted

<rec exp> received and expected data
means that the transmitted and received bytes are not equal.

9.1 Inserting Testcable 9.1

Installing the testcable may cause a strobe signal to the keyboard. If this happens, it will stop the testsystem (it is recognized with the state: stopped) and it is necessary to start the test up again with the keyboard.

10. SIO TEST 6 10.

The SIO test is testing the modem signals and data transports on both SIO channels.

Test message (9th and 10th line on display):

SIO-test: <channel identifications>: <OK or error message>
<channel identification> <baudrate>

<channel identification>
Identifies the message as belonging to a specific channel; shown as soon as the channel has completed its transfer.

Channels are: line 1:, line 2:.

<error message>
The error messages are explained in the sections 10.1 and 10.2.

<baud rate>
Explained in section 10.3.

10.1 Modem Signal Response 10.1

For the LINE I and LINE II SIO the responses of DTR and RTS are tested on the DCD and CTS pins. Loop plugs as shown in fig. 2 must be installed on the terminal LINE connections.

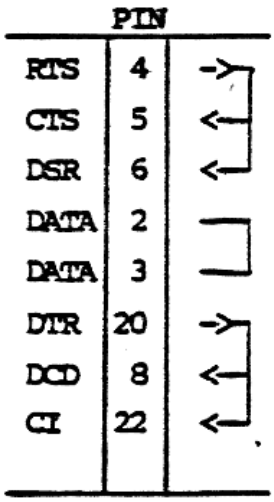

Figure 2: LINE test loop plug (CBL998).

If an error occurs on the modem signals CTS or DCD the received hex. value should be interpreted as shown in fig. 3.

Error messages from this part of the test:

CTS or DCD error, exp: <xx> rec: <xx>
Received value differs from expected; cf. fig. 3.

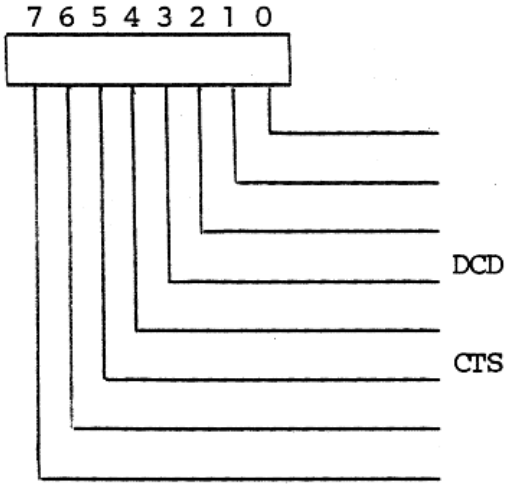

Figure 3: Modem signal response.

10.2 SIO Channels Test 10.2

When the modem signal responses have been checked, testing of LINE I and LINE II SIO channels is initiated.

The test procedure is as follows: each of the two channels is started and the control is handed over to the interrupt system and a timeout loop.

The LINE I and LINE II SIO channels are initiated to 9600 bps by the CTC.

Both channels transmit a databuffer of 4 k bytes with a counting pattern (00 FF FE FD etc.). The test loop will check the received buffers as soon as the patterns have been transferred. It also monitors both channels for timeout.

Error messages from the SIO channel test:

illegal interrupt, port: <xx>

An interrupt has occurred from a device that was not intended to interrupt, or the SIO has given a status interrupt (any change on the modem signals during data transfer will respond with this message).

parity error

A special receive interrupt with parity bit has occured.

receiver overrun

A special receive interrupt with overrun bit has occured.

data error, byte no: <xx xx> exp: <xx> rec: <xx>

The received buffer does not contain the expected pattern.

10.3 Baud Rate Measure 10.3

While the LINE I and LINE II SIO are active, their respective baud rates will be measured. This is done over a period of 1 sec. by having the CTC channel 3 giving interrupt every (approx.) 16 msec. The number of bytes received in 1 sec. is multiplied by 11 and written (8 data bits, 1 start, 1 stop and 1 parity bit).

Beware that when one channel has finished its transfer, interrupts are disabled while the buffer is checked. This means that other channels' baud rates can be a little less than else.

11. FDD TEST 7 11.

This test is testing the flexible disk controller and up to four connected drives, either 8" or 5 1/4". It is not a complete test of the flexible disk, but rather a fast verification of the basic functions of the controller and the drives connected to it.

This test is a very tough test of the drive, and errors normally appear. To check the drive use the FDD RELIABILITY TEST (see chapter 12).

If the system has not been informed that a keyboard is connected, this test is a part of the big sequential loop. If the system has been informed that a keyboard is connected, this test has to be selected by its number (number 7); that is, the big sequential loop does not involve this test.

Please note that before the FDD test is entered, a writeable diskette should be placed in the drive. For diskette stations connected to an RC700 Piccolo, it is recommendable only to use properly formatted diskettes of the type dual head, double side, soft sector, double density with the format 15 sectors/512 bytes on 8" diskettes and 9 sectors/512 bytes on 5 1/4" diskettes.

The test will initialize the controller to a step rate time of maximum 20 ms., a head unload time of maximum 160 ms. and a head load time of maximum 40 ms.

When the testing is initiated, the units which are ready, will be recalibrated. If no units are ready, the text <* all drives: not ready> will be written, and the test enters an idle state. Whenever a unit changes its state from not ready to ready, the testing will start on this unit.

If minidrives are used, the test will use some seconds to find out the drives which are ready.

Because of the READY signal when the drives are 5 1/4" diskette drives, they are only asked one time, if they are ready or not. This is done by trying a recalibration of the drives. Once a drive has been known ready or not, the testprogram expects it to be in this state throughout the test, also in looping. Every drive connected to the system is tested.

The testing sequence for a unit is first a recalibration and then the cyclic sequence, seek, write, read. Not the complete diskette is tested, but only the following 16 tracks are used for data recovery checkout:

For 8" units: 5, 60, 6, 59, 7, 58, 8, 40, 57, 9, 56, 36, 61, 37, 38, 39.

For 5 1/4" units: 5, 32, 6, 30, 7, 28, 8, 15, 31, 9, 27, 16, 29, 17, 25, 24.

The sector number varies from 1 throughout 9. On the first track the data checkout will be performed on sector 1, on the second track on sector 2 and so on.

The test pattern written on the diskette is a counting pattern of 512 bytes, which is transferred via the DMA controller channel 1. When the data is read from the diskette, it is placed in a buffer in the memory, and the write- and read-buffers are compared.

When the test is in loop mode, each byte of the test buffer is incremented by one for each pass.

If an error is detected one of the following errortexts till be written.

11.1 FDD Test Error Messages 11.1

<*fault in main status reg>

Indicates an error in the controller main status register bit 7 or 6.

<not ready>

Indicates that the ready state for a specified drive has been set to not ready (bit 3 of status register 0).

<write protected>

Indicates that the write protect bit has been set (bit 1 of status register 1).

<timeout>

The specified drive has not responded to an operation with an interrupt within approx. 2 s.

<*fault in fdc xx>

Indicates that the status bits of status register 0 are in an invalid state. xx is a hexadecimal number showing the contents of this register.

<seek error>

The drive could not find the specified track of the command issued (bit 5 of status register 0).

<command abort>

The command issued was invalid. The command was never started. It will appear if an earlier command was not terminated correctly (bit 7 = 1 and bit 6 = 0 of status register 0).

<door open>

The drive door has been opened during execution.

<recalibrate error>

The seek end bit did not occur after a recalibrate command (bit 5 of status register 0).

<track 0 signal not found>

The track 0 signal fails to occur after 77 step pulses (bit 4 of status register 0).

<missing address mark in datafield>

No address mark in the datafield (bit 0 of status register 2).

<missing address mark in id-field>

No address mark is detected in the id field (bit 0 of status register 1).

<bad cylinder>

The contents of the cylinder number on the medium are different from the internal register and cylinder number appears to be FF Hex (bit 1 of status register 2).

<wrong cylinder>

The contents of the cylinder number on the medium are different from the internal register (bit 4 of status register 2).

<cannot find sector>

The controller cannot find the sector specified in the internal register (bit 2 of status register 1).

<crc fault in id-field>

The CRC check discovered an error in the id field (bit 5 of status register 1, when bit 5 of status register 2 is zero).

<crc fault in data field>

The CRC check discovered an error in the data field (bit 5 of both status register 1 and 2).

<overrun>

If the controller is not services by the DMA controller within a certain time interval, the error occurs (bit 4 of status register 1).

<access beyond last sector>

The controller has tried to access a sector beyond the final sector of a cylinder (bit 7 of status register 1).

When one of the mentioned error messages, which has relation to a specific drive occurs, a heading is written before the message identifying the drive. The drives are numbered 0 through 3.

Texts will be written on the eighth line of the display.

12. WDC TEST 8 12.

This test is testing the WDC controller and up to two connected Winchester drives. It is not a complete test of the drives but rather a fast verification of the basic functions of the controller and the drives connected to it.

The test is only run if a controller is installed. This is signalled to the program by setting the switch S6 on the MIC board.

If the system has not been informed that a keyboard is installed, this test will be a part of the big sequential loop if the switch S6 is set.

If the system has been informed that a keyboard is connected, this test has to be selected by its number (number 8). Again, the test is only ran if switch S6 is set.

The system can be forced to perform the winchester test by pressing number 9, no matter what the switch is.

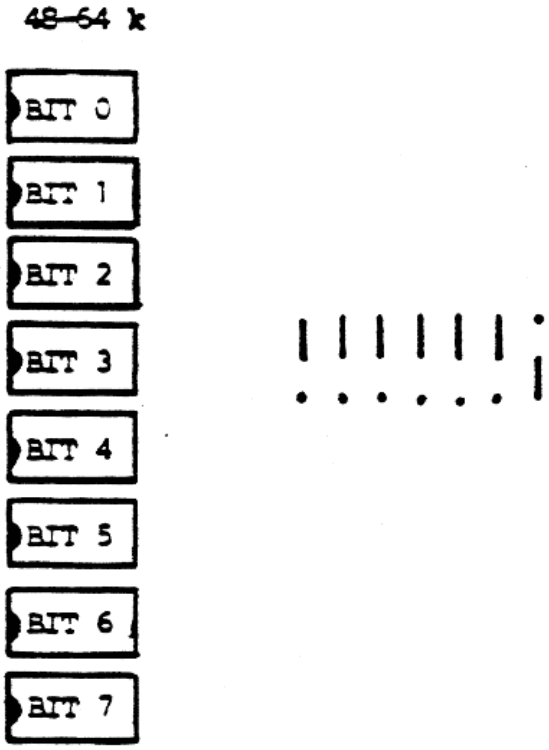

Figure 4: The switches 0-6 (here switch 6 is set).

The test will start with drive 0 and after finishing all the operation the program will check if there is a drive 1 connected, and in that case repeat the operations on drive 1.

The wdc-test starts with a test of the databus to the wdc. The test writes a register and reads it again. Only if the transmitted and the received bytes are equal the test starts. Otherwise the received and the expected bytes are written on the screen and the wdc-test is aborted.

The testing sequence for a unit is first a restore and then the cyclic sequence, format, seek, write, read. The whole Winchester disk is not tested but only cylinder 0 and 1. The system tests the winchester from head number 2 until head number 6. This will avoid destruction of the configurationsector. A track contains 17 sectors and a sector contains 512 bytes. It is formatted with an interleave rate at 4:1 and no bad sectors.

The testpattern written is a counting pattern from 00 to FF twice.
The data are transferred via the DMA controller channel 0. In testing the CRC circuit the data read from the disk are placed in a buffer in the memory (E200-E3FF) and is compared with the writebuffer (E000-E1FF).

12.1 WDC Test Error Messages 12.1

<restore error>

Indicates that the error bit in the status register is set.

After a RESTORE command, and the "track 0 error" bit in the errorregister is not set.

<track 0 error>

The track 0 error bit is set during a RESTORE command if, after issuing 1023 stoppulses, TRACK 000 line is not asserted by the drive.

<format error>

The error bit is set in the status register after a format operation.

<write error>

The error bit is set in the status register after a write operation, and the "aborted command" and the "id not found" bits are not set.

<aborted write command>

A valid write command is issued but cannot be executed based on the statusinformation from the drive.

<data mark not found>

Will be written after a READ SECTOR command if, after successfully identifying the ID field, the DATA ADDRESS MARK was not detected within 16 bytes of the ID field.

<aborted read command>

Indicates that a valid read command has been received that cannot be executed based on status information from the drive.

<id not found>

Indicates the ID field was not found.

<crc fault id-field>

<crc fault data field>

<timeout>

An interrupt was not detected within a specified period.

<bad block detect error>

This bit must not be set, while all of the sectors are formatted as good sectors.

<illegal interrupt, channel 3>

The floppy channel on the CTC has interrupted illegal.

<crc error>

Means that the CRC-check and the buffer check do not match.

13. FDD RELIABILITY TEST 13.

This test is not a part of the big testloop; that is, it is necessary to select the test by the testnumber A.

Because it is a reliability test, the program makes some error-statistics. The errors are divided into two groups: soft errors and hard errors.

When an error appears, it is counted as a soft error, and the program tries the operation again (up until the numbers of retries), if the same error appears in every retry, it is counted as a hard error and the soft error counter is decremented by the numbers of retries.

When the test is running (after some questions have been answered) it is possible to stop or to leave the test. If an "H" (capital letter) is pushed, the test will stop. To continue the test, press the "RETURN" key. Pressing the "F" (capital letter) will cause a return to the testrouter.

Pressing capital "R" will always restart the reliability test.

When the test is running, a screen-image with the errorpossibilities is shown.

The first line shows the name of the test, which can be either "FDD RELIABILITY TEST" or "DISKETTE TEST".

The number (0 or 1) after the error text refers to the headnumber.

The lines are as follows:

PASS:	passnumber.
RETRIES:	shows the number of retries, which has to be specified. In the same line the text "SOFT, HARD, HEAD, CYLINDER, SECTOR" is shown. In the

	"SOFT" column the soft errors are accumulated. In the "HARD" column the hard errors are accumulated. "HEAD, CYLINDER, SECTOR" these three columns show for each error, where the last error appeared.
NOT DETECT ID MARK:	the fdc cannot detect the id address mark.
NOT DETECT DATA MARK:	the fdc cannot detect the data address mark.
NOT FIND SECTOR:	the fdc cannot find the specified sector.
CRC ERROR IN ID:	the fdc has detect an CRC error in the id field.
CRC ERROR IN DATA:	the fdc has detect an CRC error in the data field.
SEEK ERROR:	incremented if seek error in every retry.
TIMEOUT:	interrupt does not occur within one second.
CRC-READING ERROR:	the CRC is not properly read.
FDC ERROR:	wrong status from the fdc.
WRITE ERROR:	the accumulated errors in write command, and where the last error appeared.
READ ERROR:	the accumulated errors in read command, and where the last error appeared.
CURRENT:	shows the current head-, cylinder and sectoraddress.

There are two possibilities with the reliability test. The two possibilities are either a drive test or a diskette test. Figure 5 shows the structure in the test.

Ten, or eleven questions have to be answered before the test starts. If an illegal answer has been given, a question-mark is responded until a legal number is given.

When a question is correctly answered, the next question appears on the screen. The first nine questions are:

1) drive (0/1)
   The drive on which the test is wanted. Push "0" or "1" to select drivenumber.

2) mini/maxi/quad (0/1/2)
   Selects the drive type. Minidrive, maxidrive or 96 TPI drive. Push "0", "1" or "2" to select drivetype.

3) single/double density (0/1)
   Single density = 128 bytes/sector.
   Double density = 512 bytes/sector.
   Push "0" or "1" to select density.

4) single/double sided (0/1)
   Push "0" or "1" to get next question.

5) from track
   Has to be answered with two digits. If the tracknumber is bigger than the maximum for the selected drivetype (question 1), the maximum is inserted. The maximum is for:

   mini: 36 (0-36: 37 tracks)
   maxi: 76 (0-76: 77 tracks)
   96TPI: 79 (0-79: 80 tracks)

6) to track
   Has to be answered with two digits. If the tracknumber is bigger than the maximum for the selected drivetype (question 1), the maximum is inserted. (See question number 5).

7) steprate ms (01-16)
   Two digits have to be typed to select the steprate for the drive. If the drive is a minidrive or a 96 TPI drive, the typed steprate will be multiplied by 2. Recommended steprates are for the RC702 minidrives: 20 ms (type 10), the RC702 maxidrive: 3 ms (type 03), the RC703 96 TPI: 6 ms (type 03).

8) retries (0-9)
   Determines how many retries which will be done if error.
   Push a digit to select the number of retries.

9) diskettetest no/yes (0/1)
   Push "1" to select the diskettetest.
   Push "0" to select the drive reliability test

13.1 Drive Reliability Test 13.1

If a reliability test of the flexible disk drive is chosen, the tenth question is:

10) format no/yes (0/1)
    Push "1" to format the diskette.
    Push "0" if no formatting is wanted.

The formatting is made due to the answers of the questions. The formatting is not limited to the interval [from track, to track], but to the whole diskette if double sided is chosen, or to the whole side if single sided is chosen.

If mini or 96 TPI drive is selected, the diskette is formatted with 16 sectors/track and 128 bytes/sector in single density. In double density the values are 10 sectors/track and 512 bytes/sector.

If maxidrive is selected, the diskette is formatted with 26 sectors/track and 128 bytes/sector in single density. In double density the values are 15 sectors/track and 512 bytes/sector.

The first operation in each pass is alternate seekings in the specified interval.

When the seekings are finished, the write and read of a sector begins. The program starts with sector 1 of the specified first track (from back).

After a write of a sector it is checked if the writeoperation was properly performed. If the write was OK, a read of the sector is performed. If the write was not, a read of the sector does not take place, but the program continues with a write of the next sector.

When the last sector in a track has been written and read (if possible), the same track is used on the other side, if double sided is chosen. When the cylinder is finished, the cylindernumber is incremented.

When the last cylinder in the specified interval is finished, the passnumber is incremented and the operations start again, beginning with seekings.

13.2 Diskettetest 13.2

If the diskettetest is chosen, the next two questions are:

10) read preformat no/yes (0/1)
    Push "1" to read the preformat.
    Push "0" if no reading of the preformat is wanted.

11) format no/yes (0/1)
    Push "1" to format the diskette
    Push "0" if no formatting is wanted.

After formatting the diskette the new format is read.

When reading the preformat and the new format, the reading starts at cylinder 1. The reading is made due to the answers of the cylinder.

The reason of reading the preformatting is to check the formatting done by the manufacturer (minidiskettes are not preformatted).

If a hard error appears when reading the formatting, the reading is stopped with the message "HARD ERROR". To continue a push of a key (not "R", "H" or "F") is required.

After action has been taken due to the answers of "read preformat" and "format", the specified interval is read. The reason that the interval is only read, is that the datapattern is not refreshed. This makes it possible to check the degradation in magnetism.

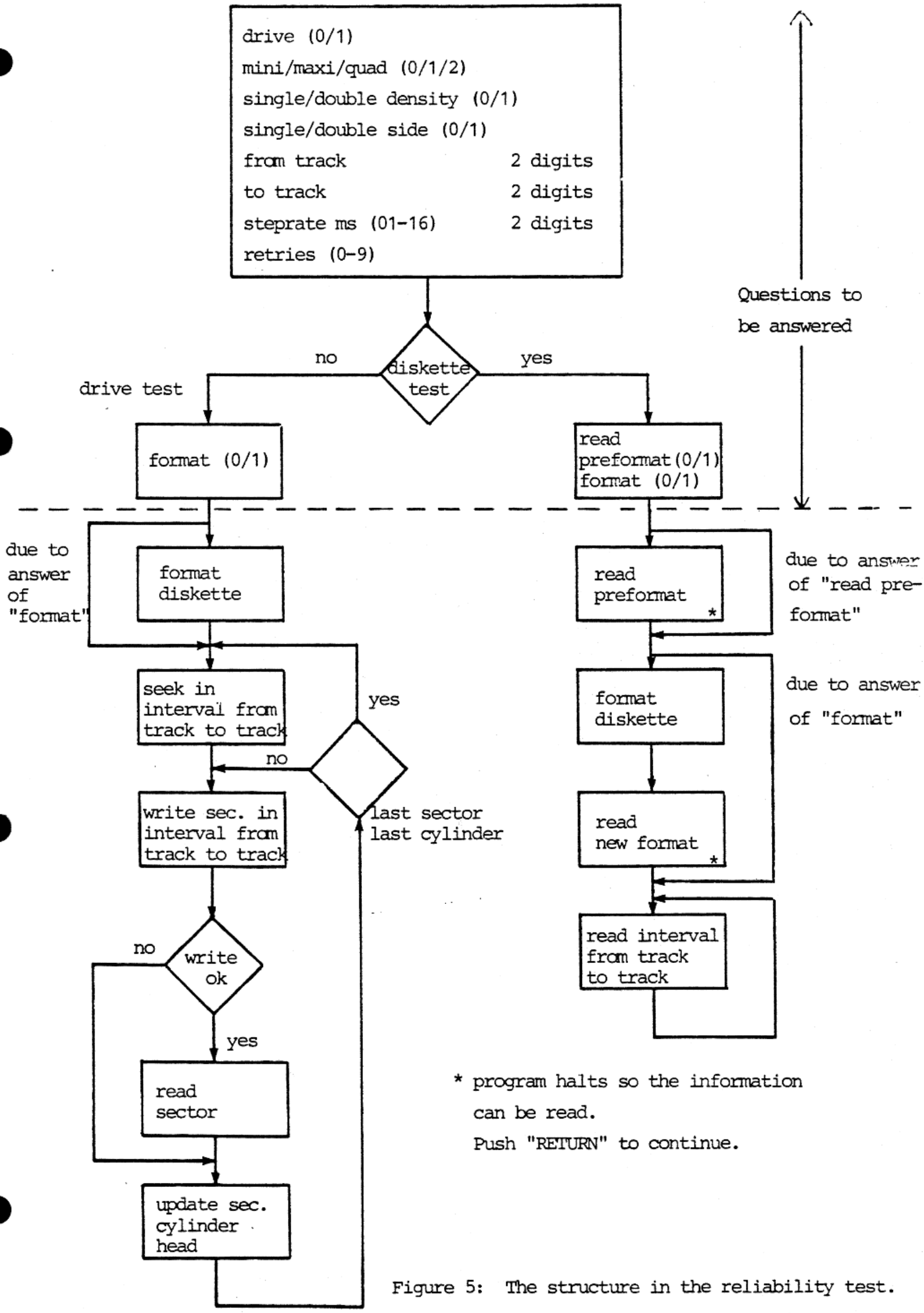

Figure 5: The structure in the reliability test.

14. CRT TEST 'B 14.

This is a visual looping test which is not a part of the test-loop but has to be selected by the testnumber B (capital letter). When the test has finished, it starts again.

When the test is running, it is possible to stop it, if the "H" key (capital letter) is pressed. To continue, press the "RETURN" key. To return to the testrouter, press "F" (capital letter).

The following should be seen:

1: The character-PROM is written for a few seconds.

2: A special code will blank the right side of the screen.

3: The character-PROM is written.

4: A special code will blank the lower half of the screen.

5: The character-PROM is written.

6: The screen is filled with "H". To adjust the screen "H" is often used.

7: Cursor test. The cursor will start in the upper left corner and move down towards the lower right corner. All four cursortypes are tested, with the character-PROM as background.

The cursor test starts with a cursor, which appears as a blinking reverse video block. The next cursortype is a blinking underline. The third cursortype is a non-blinking reverse block. The last cursor is a non-blinking underline.

8: Field attributes test. The field attributes are tested on the character-PROM. The 25 lines appear as follows:

a: 3 lines with blink
b: 3 lines with reverse video

c: 3 lines with blink and reverse video

d: 4 lines with underline

e: 3 lines with underline and blink

f: 3 lines with underline and reverse video

g: 6 lines with underline, blink and reverse video.

9: Test of the dma-channels 2 and 3 and the interrupt from the CRT controller. The first half of the screen is supplied by the dma-channel 2. An asterisk is the last character sent by this channel. The lower half is supplied by the dma-channel 3. Two asterisks are the last characters sent by this channel. After channel 3 has finished, an interruptroutine sets up the channels to screentransfers.

10: The CRT test ends up with the graphic PROM written.

After the graphic PROM is written, the test starts again with 1.

15. WDD RELIABILITY TEST C 15.

This test is not a part of the big testloop; that is, it is necessary to select the test by the testnumber C.

Because it is a reliability test, the program makes some error-statistics.

When an error appears the controller will try the operation again up until 16 times.

When the test is running (after some questions have been answered) it is possible to stop or to leave the test. If an "H" (capital letter) is pushed, the test will stop. To continue the test, press the "RETURN" key. Pressing "F" (capital letter) will cause a return to the testrouter.

Pressing "R" (capital letter) will always restart the reliability test.

When the test is running, a screen-image with the errorpossibilities is shown.

The first line shows the name of the test, which can be either "WDD RELIABILITY TEST" or "DISK TEST".

The lines are as follows:

PASS:	passnumber
DATA ADDRESS MARK NOT FOUND:	the wdc cannot detect the data address mark.
ABORTED SEEK COMMAND:	the seek command cannot be executed based on status from the drive.
ABORTED WRITE COMMAND:	the write command cannot be executed based on status from the drive.

ABORTED READ COMMAND:	the read command cannot be executed based on status from the drive.
ID NOT FOUND:	the specified id-field containing the cylinder, head and sector number was not found.
CRC ERROR IN ID:	indicates that a CRC error was encountered in an id-field.
CRC ERROR IN DATA:	indicates that a CRC error was encountered in a data-field.
CRC READING ERROR:	the CRC bytes were misread.
BAD BLOCK DETECT ERROR:	a bad block mark was erroneously read.
SEEK ERROR:	a seek error was encountered.
TIMEOUT:	a command was not completed within a specified timeinterval.
WDC ERROR:	wrong status from the wdc.
WRITE ERROR:	the accumulated errors in write command, and where the last error appeared.
READ ERROR:	the accumulated errors in read command, and where the last error appeared.
CURRENT:	shows the current head-, cylinder and sectoraddress.

There are two possibilities with the reliability test. The two

possibilities are either a drive test or a disk test. Figure 6 shows the structure in the test.

Six questions have to be answered before the test starts. If an illegal answer has been given, a question-mark is responded until a legal number is given.

When a question is correctly answered, the next question appear on the screen.

The six questions are:

1) from head
   has to be answered with one digit in the interval from 0 to 5

2) to head
   has to be answered with one digit in the interval from 0 to 5. If "from head" is bigger than "to head" the two numbers will be exchanged

3) from track
   has to be answered with 3 digits. If the tracknumber is bigger than 191 the 191 will be inserted

4) to track
   has to be answered with 3 digits. If the tracknumber is bigger than 191 the 191 will be inserted. If "from track" is bigger than "to track" the two numbers will be exchanged

5) disk test (only reading) no/yes (0/1)
   push 0 to select drive reliability test
   push 1 to select disk test

6) format no/yes (0/1)
   push 0 if no formatting is wanted
   push 1 to format
   The formatting is done due to the answers of "from head", "to head", "from track" and "to track". The formatting is performed only in the specified volume.

Remark that the formatting done with this testprogram is not compatible with the formatting done by HDFORM.

15.1 Drive Reliability Test 15.1

The operations in each pass are first write a sector, read the same sector, invert the bitpattern and write and read the sector again. The read buffer is compared with the write buffer.

15.2 Disk Test 15.2

In this part the specified interval is only read. The read data are checked for CRC error.

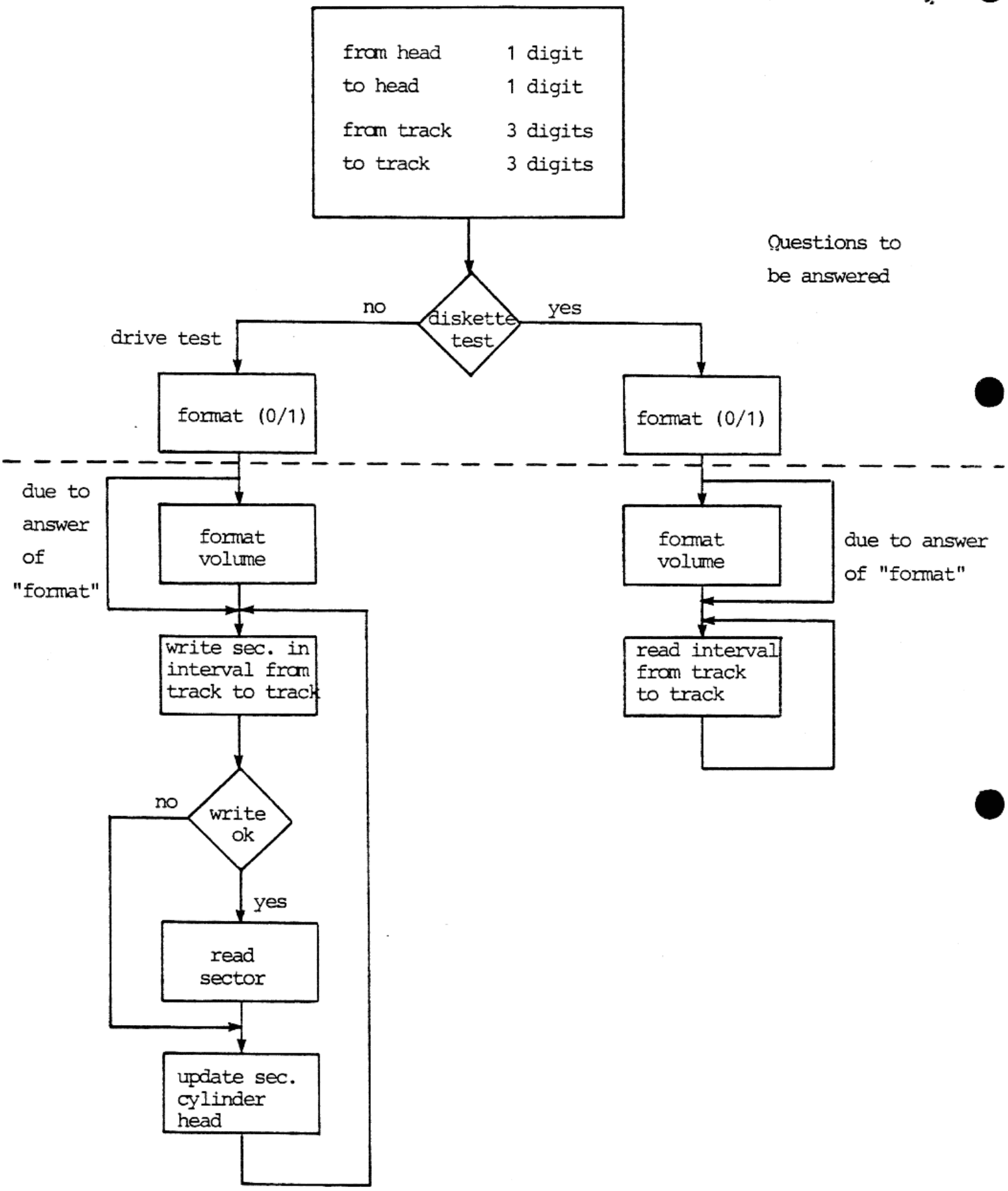

Figure 6: The structure in the reliability test.

16. ERROR CODES 16.

The testrouter outputs an error code, which is specific for the type of error discovered on port 50 Hex. This enables the use to run the testsystem on a MIC board alone without display. The error information may then be detected if a device which can decode the numbers is installed on the system bus.

The error codes are as follow:

0: OK, no error
1: PROM checksum error
2: RAM error
3: data error in DMA test
4: DMA channel 1 has not set the terminal count bit within 200 ms. in DMA test
5: not used
6: not used
7: not used
8: not used
9: not wanted interrupt in CTC test
A: not used
B: CTC test has timed out without interrupt
C: not used
D: not used
E: not used
F: not used
10: not used
11: not used
12: not used
13: not used
14: not used
15: not used
16: not used
17: data error in main memory refresh test
18: FDC not ready to receive or transmit
19: the wrong data direction in FDC
20: FDC fault in status register
31: fault in FDC (in FDD test)

32: seek error
33: FDC command abort
34: open door
35: recalibration error
36: track 0 signal not found
37: missing address mark in data field
38: bad cylinder
39: wrong cylinder
3A: missing address mark in ID field
3B: cannot find sector
3C: CRC fault in ID field
3D: CRC fault in data field
3E: drive not ready
3F: overrun in FDC
40: trying to access beyond last cylinder
41: drive is write protected
42: flexible disk drive timeout
43: flexible disk data error
50: timeout in WDC test
51: track 0 error in wdc-test
52: restore error in wdc-test
53: format error in wdc-test
54: aborted write command in wdc-test
55: id not found in wdc-test
56: CRC fault in id field in wdc-test
57: write error in wdc-test
58: CRC error in wdc-test
59: data mark not found in wdc-test
5A: aborted read command in wdc-test
5B: CRC error in datafield in wdc-test
5C: bad block detect error in wdc-test
5D: illegal interrupt in wdc-test.

RETURN LETTER

Title: Testsystem for the RC701/RC702 Version 1.1 User's Guide. RCSL No.: 44-RT2061

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